Universal voluntary HIV testing with immediate antiretroviral therapy as a strategy for elimination of HIV transmission: a mathematical model

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Summary

Background Roughly 3 million people worldwide were receiving antiretroviral therapy (ART) at the end of 2007, but an estimated 6·7 million were still in need of treatment and a further 2·7 million became infected with HIV in 2007. Prevention efforts might reduce HIV incidence but are unlikely to eliminate this disease. We investigated a theoretical strategy of universal voluntary HIV testing and immediate treatment with ART, and examined the conditions under which the HIV epidemic could be driven towards elimination.

Methods We used mathematical models to explore the effect on the case reproduction number (stochastic model) and long-term dynamics of the HIV epidemic (deterministic transmission model) of testing all people in our test-case community (aged 15 years and older) for HIV every year and starting people on ART immediately after they are diagnosed HIV positive. We used data from South Africa as the test case for a generalised epidemic, and assumed that all HIV transmission was heterosexual.

Findings The studied strategy could greatly accelerate the transition from the present endemic phase, in which most adults living with HIV are not receiving ART, to an elimination phase, in which most are on ART, within 5 years. It could reduce HIV incidence and mortality to less than one case per 1000 people per year by 2016, or within 10 years of full implementation of the strategy, and reduce the prevalence of HIV to less than 1% within 50 years. We estimate that in 2032, the yearly cost of the present strategy and the theoretical strategy would both be US$1·7 billion; however, after this time, the cost of the present strategy would continue to increase whereas that of the theoretical strategy would decrease.

Interpretation Universal voluntary HIV testing and immediate ART, combined with present prevention approaches, could have a major effect on severe generalised HIV/AIDS epidemics. This approach merits further mathematical modelling, research, and broad consultation.

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Introduction

25 years after the discovery of HIV,[1] control of the HIV epidemic remains elusive and some have called for a re-examination of the approach to control this virus.[2] Development of an effective HIV-1 vaccine remains a remote possibility, and trials of vaginal microbicides have not shown any protective benefit.[3,4] Where HIV transmission is mainly heterosexual, male circumcision can reduce adult heterosexual HIV transmission, but only by about 40% at the overall population level.[5,6] A call has been made to focus prevention interventions on high-risk populations[7] and to expand programmes for female sex workers and their clients,[8] and injecting drug users.[9] Few people are aware of their HIV status, and rapid expansion of voluntary HIV testing and counselling has been recommended by WHO.[10] About 3 million people worldwide had been given antiretroviral therapy (ART) at the end of 2007[10] but an estimated 6·7 million were still in need of it and a further 2·7 million were infected with HIV in 2007.[10–12]

At present there is inadequate evidence for WHO to provide guidance on the role of ART for people living with HIV as a strategy to prevent further sexual transmission. To control the HIV/AIDS epidemic, infectious individuals would have to be rendered non-infectious, or susceptible people protected from infection. Vertical transmission of HIV can be eliminated by testing of mothers and blocking of transmission through the use of antiretroviral drugs, accompanied by elective caesarean section and the use of replacement infant feeding.[13] Although increasing emphasis is being placed on positive prevention[14,15] and provider-initiated HIV testing and counselling,[16] no large-scale studies have been undertaken of the effect of diagnosing all HIV-positive people early and treating them immediately.

Present guidelines suggest that ART should be started when infected people reach specific immunological or clinically-defined stages of disease to keep subsequent morbidity and mortality in individual patients to a minimum.[17] Wherever ART has been implemented it has had a substantial and rapid effect on survival for individuals and within populations.[18] The effect of treatment on transmission and the possible public-health benefits have, with some exceptions, received less

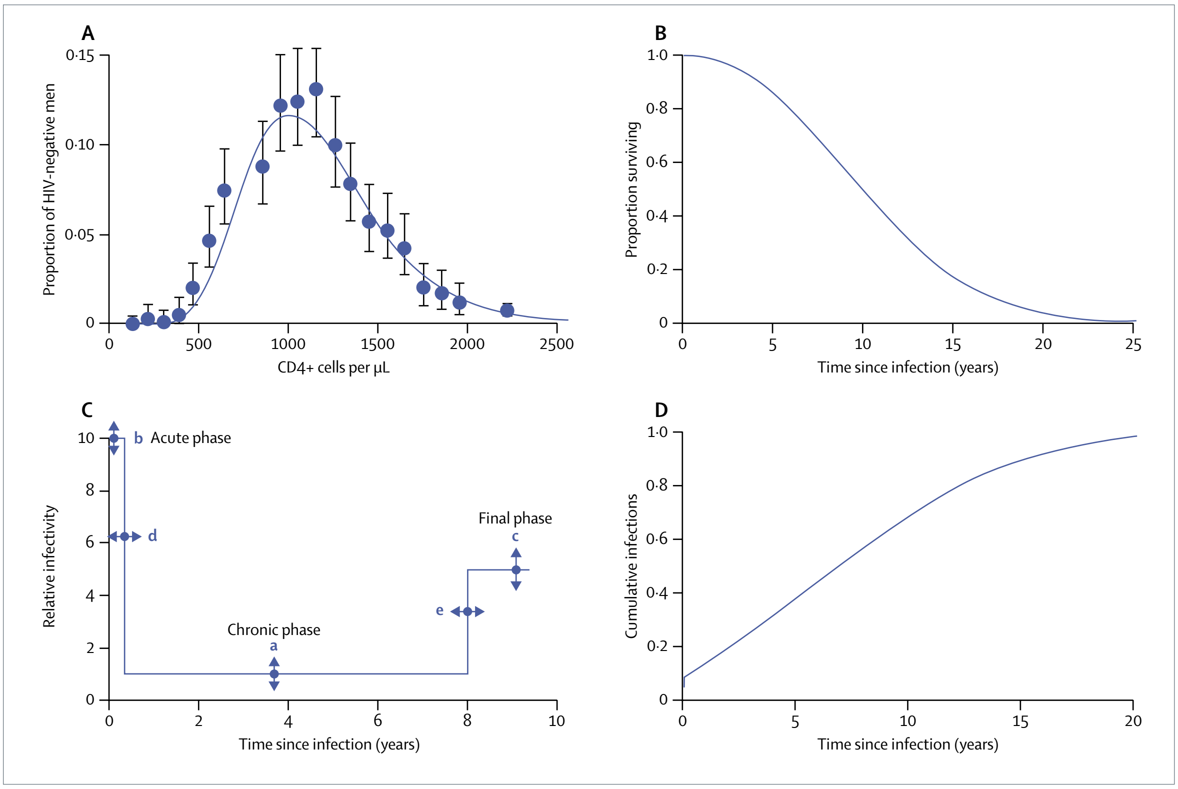

Figure 1: **Theoretical basis for the stochastic and deterministic transmission models**
(A) The observed distribution of CD4+ cells in HIV-negative men in South Africa.[29,30] Error bars indicate 95% CIs. (B) The Weibull survival curve, with HIV infection used in the model on the basis of data from the CASCADE study.[30] (C) Schematic diagram of the change in infectivity with time in a person who survives for 10 years. The infectivity during all three phases (a–c) can be varied, as can the duration of the acute and final phases (d and e). (D) The cumulative distribution of the time between the infection of the index case and the infection of a secondary case for the baseline parameters as calculated from the model. The initial rapid increase (from 0% to 10%) is the result of the relatively high infectivity during the acute phase.

attention.[19–22] The use of ART can reduce the plasma viral load by up to six orders of magnitude,[23] and several investigators have assessed the effect of ART on transmission.[19,20,24] However, the high viral load during the acute phase of infection, the long duration of infectiousness, the present policy of limiting costly and potentially toxic ART to people whose immune systems are severely compromised, and low coverage can reduce the extent to which the use of ART reduces transmission.

Despite substantial efforts to expand access to voluntary HIV testing, nearly 80% of HIV-infected adults in sub-Saharan Africa are unaware of their status and more than 90% do not know whether their partners are infected with HIV.[10,25] Present approaches to HIV testing, prevention, and treatment are unlikely to bring about a rapid reduction in HIV incidence, and demand for treatment in countries that are most heavily affected will continue to grow. Reduction in HIV incidence, with the goal of eventual elimination, would require that the case reproduction number R_0—the number of secondary infections resulting from one primary infection in an otherwise susceptible population—is reduced to and kept below 1.[26] A potential shift in strategy is to diagnose all HIV-infected people as soon as possible after infection and provide them with immediate ART. In considering the use of ART to eliminate transmission, we focused on two questions: how often would people have to be tested and how soon after testing positive should they start ART? In this hypothetical modelling exercise, we examined a strategy of universal voluntary HIV testing and immediate treatment with ART in the context of a generalised heterosexual epidemic of the same intensity as in southern Africa, and examined the conditions under which the epidemic could be driven towards elimination. The results have potential implications for HIV prevention that require broad consultation.

Methods

Study design

To establish a hypothetical HIV epidemic, we relied on available data from South Africa as the test case for a generalised HIV epidemic, representing 17% of all people living with HIV.[11] Our hypothetical test case assumed, as in South Africa, that almost all transmission was heterosexual rather than homosexual. This assumption holds true for South Africa and similar settings, and was supported by the observation that the prevalence of HIV in South African men is estimated to be 58% of the prevalence in women.[27] Furthermore, we assumed that intravenous drug use does not contribute substantially to the overall rates of infection.[28]

We used a stochastic model to explore the effect of various treatment strategies and model parameters on R_0, allowing transmission to vary between the acute and chronic phase of HIV infection. To set an elimination target for the purposes of this study, we defined HIV elimination as a reduction in incidence to less than one case per 1000 people per year. We used a deterministic transmission model to explore the effect of various HIV testing and treatment strategies on the long-term dynamics of the epidemic. The models were programmed in Visual Basic (Microsoft 2002).

Stochastic model

In the stochastic model, we chose an initial value for a person's CD4+ cell count and a survival time, which were taken from population distributions derived from available data. For the pre-infection distribution of CD4+ cell counts (figure 1A) we used data from a survey in Orange Farm, South Africa.[29] We assumed that the CD4+ cell count decreased by 25% immediately after infection and linearly thereafter.[29] Every month, the index case could infect another person, be tested for HIV, or, if HIV positive, start treatment at a preset CD4+ cell count.

We assumed that survival after infection with HIV, without ART, would follow a Weibull distribution[29] (figure 1B) with a mean of 11 (SD 0·5) years.[31] The acute phase would last for 2 months, during which time the infectivity would be ten times higher than in the chronic phase (figure 1C).[32,33] The final phase would last for 5% of the survival time without ART, during which time the infectivity would be five times higher than in the chronic phase.[33] If a person was tested and if their CD4+ cell count was less than a specific threshold, they would be given ART. Once receiving ART, we assumed that their infectiousness fell to 1% of their value before treatment on the basis of the relation between plasma viral load and ART and estimated decreases in viral load for people receiving ART.[23,34]

We started the model without ART by adjusting the chronic-phase infectivity to obtain a value of R_0 equal to 7 and a doubling time of 1·25 years to match the data describing the epidemic trend in South Africa.[29] The parameters in figure 1C could all be varied to examine the effect of introducing ART at various CD4+ cell counts on the values of R_0, to calculate the time from the infection of the index case to the infection of each secondary case, and to assess the cumulative number of infections over time (figure 1D). The model would provide the mean and frequency distribution of each of these statistics, which were obtained over many runs (figure 1D).

Deterministic transmission model

The deterministic transmission model in figure 2 had four compartments for people infected with HIV to give a close fit to the observed Weibull survival distribution (figure 1B). The compartments in the transmission model simulated the progression from infection to AIDS (and were unrelated to the acute and final phases in the stochastic model).

To calibrate the model we used the prevalence of HIV in South African adults aged 15 years and older, which

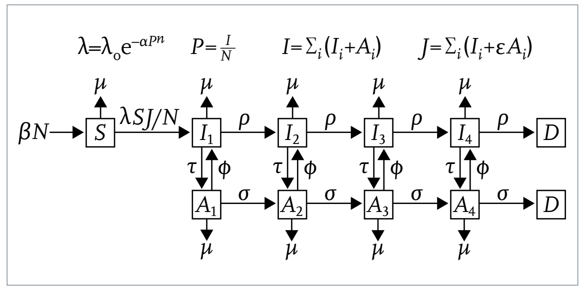

Figure 2: **Transmission model for HIV infection and antiretroviral therapy (ART) provision**
N represents population aged 15 years and above. People enter into the susceptible class (*S*) at a rate βN, become infected at a rate $\lambda SJ/N$, progress through four stages of HIV (I_i, i=1–4) at a rate ρ between each stage, and then die (D). The background mortality rate is μ and people are tested at a rate τ. If they are tested and put onto ART, they move to the corresponding ART box A_i (i=1–4), where they progress through four stages at a rate σ and then die. The term governing transmission contains the factor $J \propto (I_i+\varepsilon A_i)$ where ε allows for the fact that people receiving ART are less infectious than are those who are not. They might also stop treatment or the treatment might become ineffective, in which case they return to the corresponding non-ART state at a rate ϕ. To allow for heterogeneity in sexual behaviour and for the observed steady state prevalence of HIV, we let the transmission decrease with the prevalence, *P*. If n=1, the decrease is exponential; if $n=\infty$, the decrease is a step function. Both have been used in previous models.[5,29]

For more on **both models** see http://www.who.int/hiv/topics/treatmentasprevention

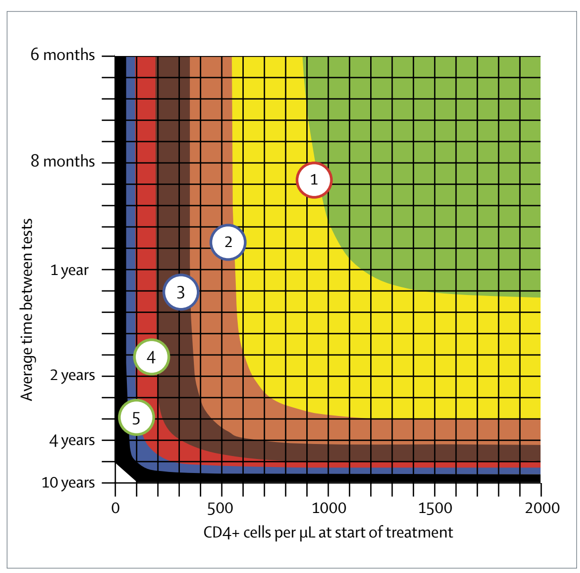

Figure 3: **Relation between HIV testing frequency, CD4+ cell count, and R_0**
R_0 (the number of secondary infections resulting from one primary infection in an otherwise susceptible population) plotted against the CD4+ cell count at which treatment starts for different frequencies of HIV testing (average time between HIV tests represented in years and months). Numbers in circles represent R_0 values. Green shading: R_0<1; yellow: 1<R_0<2; orange: 2<R_0< 3; brown: 3<R_0< 4; red: 4<R_0<5; blue: 5<R_0<6; and black: 6<R_0< 7.

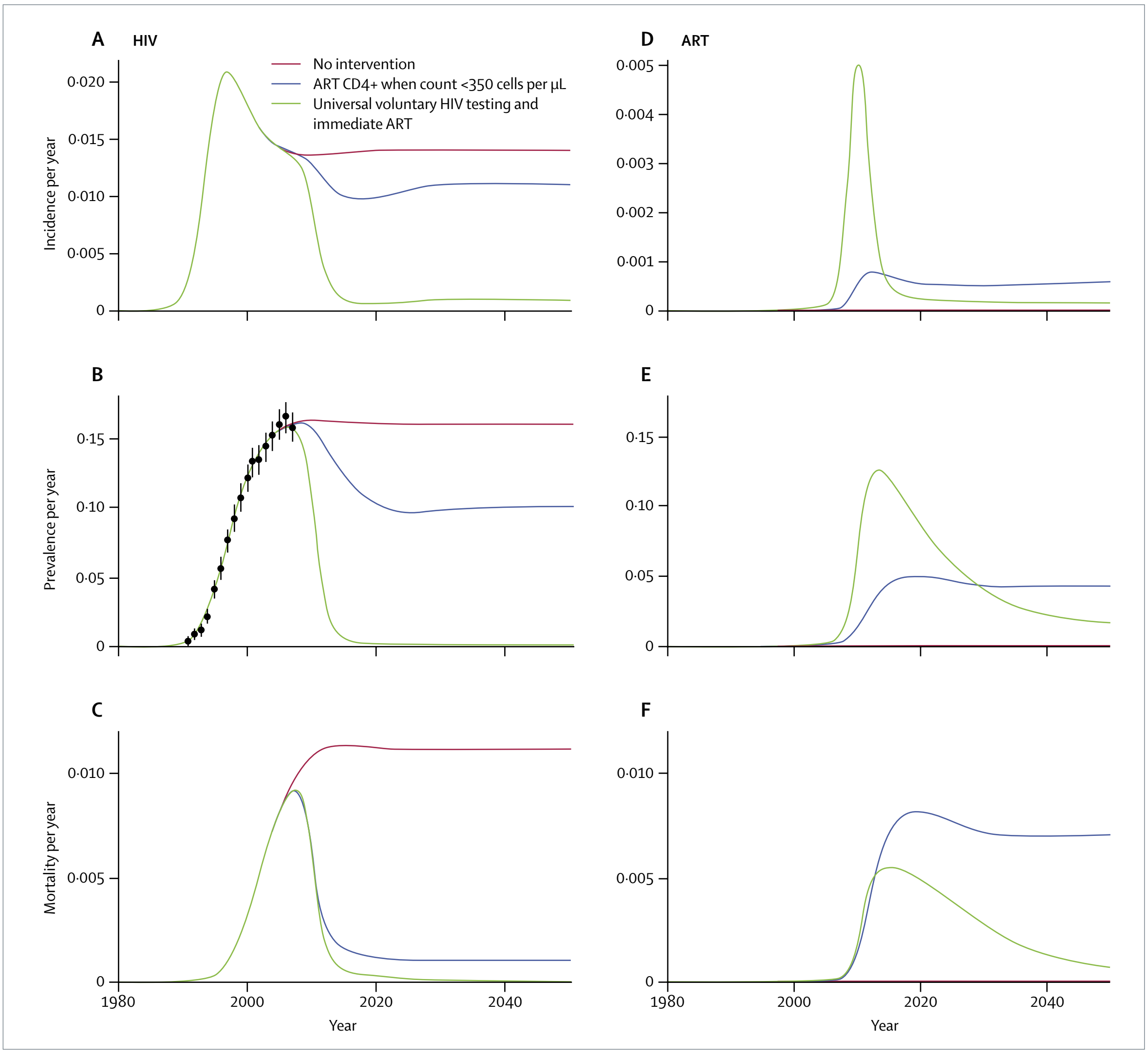

Figure 4: **HIV incidence (A), prevalence (B), and mortality (C) and the incidence (D), prevalence (E), and mortality (F) of people placed on antiretroviral therapy (ART)**

Data are proportions of adolescents and adults aged 15 years and older. Left panels, A–C, show HIV incidence, HIV prevalence, and mortality under the three scenarios. In B, data points with error bars give estimated prevalence of infection in adults derived from South Africa antenatal clinic data.[35] Right panels, D–F, show incidence of ART—ie, the rate at which people start ART—the proportion of people receiving ART, and the mortality in those receiving ART.

was obtained by scaling data from the national antenatal clinic survey[35] to the UNAIDS estimate for adults in 2005.[36] To obtain the best fit we varied the timing of the epidemic, the transmission parameter (λ), and the parameters that account for heterogeneity in sexual behaviour (α and n) (figure 2).

A proportion of people offered ART might refuse treatment or fail treatment within the first few weeks; thus a proportion of people in the model did not start treatment. To allow for subsequent drop out because of logistical and other challenges such as adherence to treatment, toxic effects of drugs, and drug resistance, individuals stopped taking ART at a rate ϕ per head per year (figure 2). We used failure, refusal, and retention rates that have been achieved in the national programme in Malawi.[37] Data from the national ART programme in Malawi[37] suggest that up to 8% of people drop out immediately or soon after starting treatment and then at between 1% and 3% per year, excluding those who die. We assumed a long-term drop-out rate of 1·5% per year. We assumed that first-line treatment would fail in 3% of individuals every year and that those individuals would be identified and put on second-line treatment. Although their prognosis on second-line drugs would then be the same as for those on first-line drugs, this assumption has important implications for the cost of treatment. When we introduced the intervention, we assumed that coverage would increase logistically to 50% by 2012 and 90% by 2016.

We examined a strategy of universal voluntary HIV testing and immediate ART in which all adults in our

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test-case community accepted being tested for HIV once a year on average, and all HIV-infected people started ART as soon as they were diagnosed HIV positive (ie, irrespective of clinical stage or CD4+ cell count). The sensitivity and specificity of the present generation of HIV tests are near 100% and result in a high predictive value, especially when different tests are used together as part of a quality-controlled algorithm.[38,39] Therefore, we assumed 100% sensitivity and specificity for the modelling exercise. We compared the theoretical strategy of universal voluntary HIV testing and immediate ART with a strategy in which ART was started with a CD4+ count less than 350 cells per μL on the basis of optimum implementation of present guidelines from the Southern African Clinicians Society[40] in which we assumed that all infected individuals would have presented to a health service before their CD4+ count fell to 350 cells per μL and would start ART at that threshold. For this initial comparison, we focused on the effect of ART and excluded the effect of other prevention interventions. We used the deterministic transmission model to compare the effect of the two interventions on the incidence, prevalence, and mortality of HIV-positive people who would be, and would not be, receiving ART.

We then used the deterministic model to examine the time to elimination after implementation of the theoretical strategy. We combined yearly universal voluntary HIV testing and immediate ART for all people who tested HIV positive irrespective of CD4+ cell count or clinical stage with a full package of relevant adult prevention interventions. Particularly in the early stages of the programme, this strategy could reach and place on ART people who would not have been tested for HIV and those with CD4+ counts greater than 350 cells per μL. We also assumed that other adult interventions for prevention of HIV would contribute to reductions in transmission. These interventions could include male circumcision, behaviour-change programmes, condom promotion, and treatment of curable sexually transmitted infections. We assumed that these other interventions together would reduce transmission by 40% and would be rolled out at the same rate as the ART programmes.

	Deaths (thousands)	Deaths averted (thousands)
Neither strategy		
2015	269	..
2030	263	..
2050	263	..
2008–50	11 078	..
ART started when CD4+ count <350 cells per μL		
2015	193	76
2030	202	61
2050	210	53
2008–50	8658	2419
ART started when CD4+ count <350 cells per μL and universal voluntary HIV testing/immediate ART		
2015	165	104
2030	76	187
2050	17	246
2008–50	3879	7199
ART started when CD4+ count <350 cells per μL, universal voluntary HIV testing/immediate ART, and other adult prevention strategies		
2015	164	105
2030	72	191
2050	12	251
2008–50	3727	7350

ART=antiretroviral therapy.

***Table:* Estimated number of AIDS-related deaths for the years 2015, 2030, 2050, and 2008–50 with different strategies**

Cost analysis

To estimate the funding needed to implement the two strategies in our test-case scenario of a severe generalised epidemic, we assumed a cost for first-line drugs—including drug delivery, HIV and laboratory testing, and patient management—of US$727 (range $290–1163) for first-line drugs and $3290 ($2497–4083) for second-line drugs.[41–43] We assumed that 30% of treatment costs were for antiretroviral drugs[43] and that 3% of people on first-line drugs would fail and need second-line drugs every year. We also incorporated available data suggesting that costs for HIV testing range from 0·5–3% of the per person first-line ART costs and 0·1–0·6% of the second-line ART costs (Mermin J, Centers for Disease Control and Prevention Kenya, Coordinating Office for Global Health, CDC, Nairobi, Kenya, personal communication). To compare the cost of the theoretical strategy for the hypothetical case, we assumed that 17% or $2·87 billion of estimated yearly global funding for HIV/AIDS up to 2008, and 17% or $8·84 billion of UNAIDS projected yearly needs for universal access up to 2015, would be available for HIV/AIDS control as described for South Africa.[36,44] UNAIDS universal-access estimates for prevention, care, and treatment include treatment for 13·7 million people by 2010.[35,44] Cost calculation results apply to this hypothetical scenario analogous to South Africa, unless otherwise specified.

Role of the funding source

There was no funding source for this study. All authors had access to the data and agreed to submit for publication.

Results

Figure 3 shows results from the stochastic model as contours of R_0 plotted against the CD4+ cell count at which ART would start and the frequency of testing. To reduce R_0 to less than 1 (green area), adolescents and adults would need to be tested at least once per year and started on ART when their CD4+ count is greater than

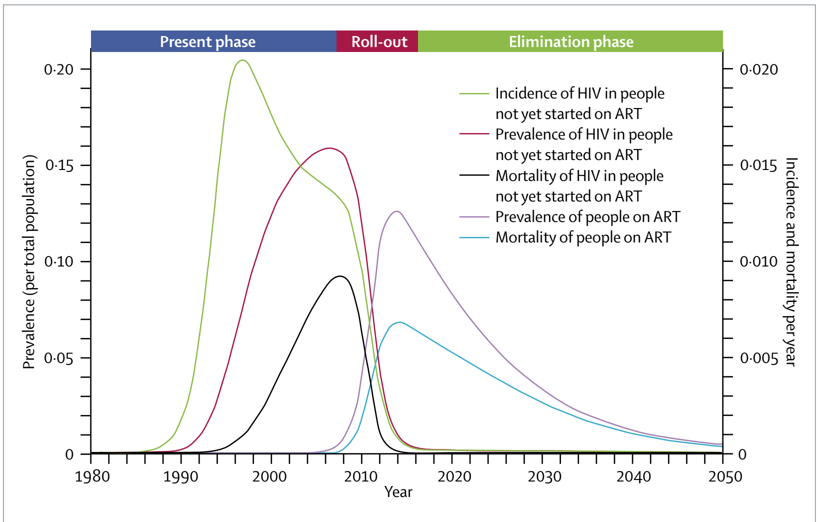

Figure 5: **Time trends resulting from application of universal voluntary HIV testing and immediate ART strategy for people who test HIV positive, in combination with other adult prevention interventions that reduce incidence by 40%**

The programme implementation start date is arbitrarily set as immediate, with coverage increasing logistically to 50% by 2012 and 90% by 2016. The parameters are τ=1·0 per year; ξ=0·08; ϕ=0·015 per year; and η=0·10. See figure 2 legend for description of these variables.

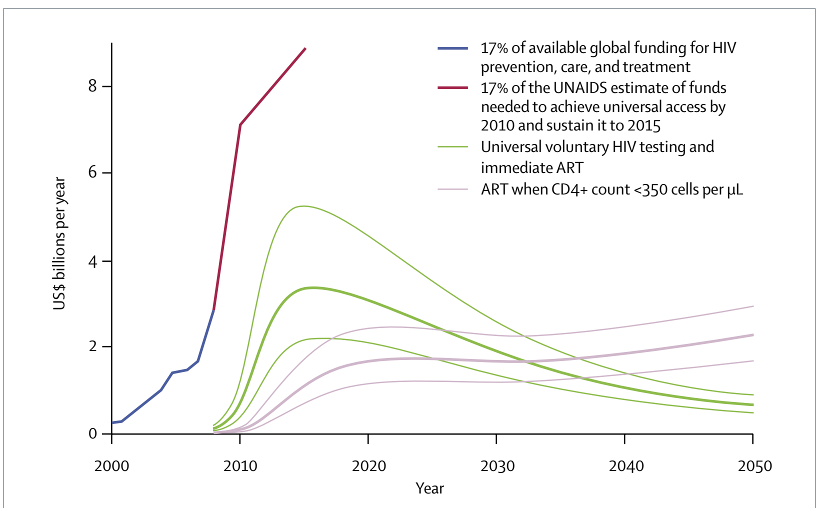

Figure 6: **Yearly cost of the two strategies compared with available and projected funding for HIV/AIDS for the test-case country**

Heavy lines correspond to expected treatment costs, light lines to high and low estimates of treatment costs for the hypothetical epidemic representing 17% of worldwide HIV prevalence, as described in the methods section. Blue line: 17% of available global funding for HIV prevention, care and treatment;[44] brown line: 17% of the UNAIDS estimate of funds needed to achieve universal access by 2010 and to sustain it to 2015.[36]

900 cell per µL. In South Africa, the average value of the CD4+ count immediately after seroconversion is about 884 cells per µL,[29] so most adolescents and adults would need to start ART as soon as they are diagnosed with HIV to ensure that R_0 stays below 1. Figure 3 also shows that if adolescents and adults are tested on average once a year, starting ART at a CD4+ count of 200 cells per µL could reduce R_0 to 4, starting at 350 cells per µL could reduce it to 3, and starting at 500 cells per µL could reduce it to 2·5.

Although these other strategies could have a substantial effect on transmission, morbidity, and mortality, and are in agreement with previous studies,[22,24,45] they would not be sufficient to reduce R_0 to less than 1 and move the epidemic towards elimination. The stochastic model shows that testing all adolescents and adults at least 15 years old once a year, on average, and starting individuals on ART as soon as they test positive for HIV would reduce R_0 below 1 and eventually eliminate HIV. The model allows for a high level of concurrency and for a much higher infectiousness during the acute phase than during the chronic phase.[46]

The strategy of starting ART when CD4+ count is less than 350 cells per µL approximates the optimum implementation of many national guidelines for HIV treatment, including those recommended for South Africa.[40] In countries most heavily affected by HIV, most people are not aware of their HIV status,[10] and the median CD4+ count when individuals start ART is often much lower than 350 cells per µL.[47] Figure 4 shows the deterministic transmission model of the two strategies—universal HIV testing and immediate ART strategy versus starting ART when the CD4+ count is less than 350 cells per µL. The strategy of starting ART when CD4+ count is less than 350 cells per µL could reduce both HIV incidence (figure 4A) and prevalence (figure 4B) by roughly 30% but could give a much greater reduction in mortality (figure 4C) since everyone, apart from those who refuse treatment, could start ART when their CD4+ cell count reached this level.

The rate at which adolescents and adults start ART could increase to about 0·08% per year by 2014 (figure 4D), the prevalent number receiving ART could increase to about 5% of the adult population by 2020 (figure 4E), and the mortality of people receiving ART could increase to about 0·8% per year by 2020 (figure 4F). By contrast, the theoretical strategy of yearly universal voluntary HIV testing and immediate ART could reduce HIV incidence, prevalence, and mortality to about one case per five thousand adolescents and adults per year, since by 2016 most infected people could be receiving ART (figure 4). The number of people starting ART and the number already receiving ART could initially be much greater than with the strategy of starting ART when CD4+ count is less than 350 cells per µL (figure 4), but values decrease rapidly as transmission is interrupted. By 2016, fewer people could be starting ART and by 2030 fewer people could be receiving ART than with the strategy of starting ART when CD4+ count is less than 350 cells per µL (figure 4). Mortality could be much the same for both strategies until 2016, when the number of deaths per year could fall rapidly with the theoretical strategy of universal voluntary HIV testing and immediate ART (figure 4).

We estimated a substantial difference in number of lives saved between the strategies (table). The number of HIV deaths between 2008 and 2050 could be about

11 million without ART, about 8·7 million with the strategy of starting ART when CD4+ count is less than 350 cells per µL, and 3·9 million with the theoretical strategy of universal voluntary HIV testing and immediate ART combined with the CD4+ count less than 350 cells per µL strategy. The combined strategy could reduce the number of HIV deaths up to 2050 by 55% compared with the strategy of starting ART when CD4+ count is less than 350 cells per µL, at which time the mortality could be only 0·04% per year and falling, rather than 0·6% per year and rising.

Figure 5 shows the time to elimination after implementation of the theoretical strategy in combination with other adult prevention interventions. The implementation start date of the programme is arbitrary and the time to elimination is dependent on achievement of programme scale-up. With the assumption of an immediate start date and full programme scale-up by 2016, HIV transmission could switch from the present endemic phase, in which most people living with HIV would not be receiving ART, to the elimination phase, in which most would be receiving ART, at around 2010. By 2020, mortality in people not receiving ART could have fallen to about two per thousand adolescents and adults per year. Once the epidemic is in the elimination phase, the focus of control efforts could also change. Before 2010, the focus could be on ensuring that HIV testing is widely done and ART is immediately available. After 2010, there could be an increasing need to ensure that people who are now receiving ART are fully adherent, that switching to second-line therapy is prompt and efficient, that sexual partners, in particular, are monitored for evidence of secondary HIV infections, and that population-based drug resistance is monitored.[48]

We used the model to compare the cost of the theoretical strategy of universal voluntary HIV testing and immediate ART and the strategy of starting ART when CD4+ count is less than 350 cells per µL (figure 6). Initially, the cost of the universal voluntary HIV testing and immediate ART strategy is greatest—in 2015, the cost of this theoretical strategy is almost three times more than that of the strategy of starting ART when CD4+ count is less than 350 cells per µL. However, after 2015, the yearly cost of the strategy of universal voluntary HIV testing and immediate ART falls and is less than the cost of the strategy of starting ART when CD4+ count is less than 350 cells per µL after 2030. As the yearly costs for the strategy of universal voluntary HIV testing and immediate ART fall, the costs for the strategy of starting ART when CD4+ count is less than 350 cells per µL could continue to rise as more people need ART.

The funding needed to implement the theoretical strategy for an epidemic of South African-type severity peaks in 2015 at \$3·4 billion per year (range \$2·2 billion–\$5·3 billion). Although the initial yearly cost of the theoretical strategy is higher than the present strategy, it is well within UNAIDS projections of the \$8·84 billion needed every year for universal access to prevention, care, and treatment in a South African-type situation in 2015. In the long term, costs could reduce to very low amounts as progress towards elimination is achieved.

Discussion

The results show that universal voluntary HIV testing once a year of all people older than 15 years, combined with immediate ART after diagnosis, could bring about a phase change in the nature of the epidemic. Instead of dealing with the constant pressure of newly infected people, mortality could decrease rapidly and the epidemic could begin to resemble a concentrated epidemic with particular populations remaining at risk. The focus of control would switch from making ART available to people with greatest need to providing support and services for those who are receiving ART. Transmission could be reduced to low levels and the epidemic could go into a steady decrease towards elimination as those receiving ART grow older and die.

Although other prevention interventions, alone or in combination, could substantially reduce HIV incidence, our model suggests that only universal voluntary HIV testing and immediate initiation of ART could reduce transmission to the point at which elimination might be feasible by 2020 for a generalised epidemic, such as that in South Africa. This analysis lends support to, and extends, earlier analyses suggesting that rapid scale-up of conventional ART approaches could greatly reduce mortality[49] and have a substantial effect on HIV incidence.[19,20] However, other studies that examined scaling up ART on the basis of present HIV testing access and clinical eligibility criteria for ART did not show the reduction of R_0 to less than 1, which could be seen with the theoretical strategy of universal voluntary HIV testing and immediate ART.[22,45]

The main restrictions in this modelling exercise relate to need for much better data, especially for programmatic aspects of the intervention. Better data are needed for the acceptability and uptake of universal voluntary HIV testing, the infectiousness of people receiving ART, adherence, behaviour change after starting ART, and rates of emergence of resistance. Although we undertook a preliminary costing exercise, a full economic analysis of the proposed strategy could further improve our understanding of the economic implications of the theoretical strategy. Several trials are in progress, and many of these data could become available. A trial of our theoretical strategy is technically feasible, especially in view of the expected rapid effect on incidence. Such a trial would address the assumptions in the model concerning practical aspects of implementing the strategy.[50,51]

The studied strategy would pose implementation challenges. It could initially be costly and labour intensive, and difficulties could arise from drug resistance or adverse events related to medication. It would be necessary to provide consistent, secure access to HIV rapid-test kits and first-line and second-line antiretroviral drugs, to ensure high levels of adherence, and to monitor the programme carefully, especially when most people with HIV are receiving ART. The sensitivity and specificity of the present generation of HIV tests are near 100% and result in a high positive predictive value, particularly when different tests are used together as part of a quality-controlled algorithm. However, the scale-up of HIV testing to millions of people means that even a small false-positivity rate could lead to many people being falsely diagnosed as HIV positive.[38,39] Similar to HIV testing efforts at present, this theoretical strategy will require substantial attention to the testing algorithm and quality assurance. The behavioural implications of a large cohort of people receiving ART on the community are largely unknown, and particular attention would have to be given to the sexual partners of people on this treatment. We have assumed a moderately paced scale-up that reaches full coverage by 2016, and we used rates of failure, refusal, and retention that have been achieved in the national programme in Malawi.[37] If the intervention is scaled up more slowly or started at a later date, the long-term effect would be the same but delayed.

Universal access implies universal knowledge of serostatus, but even without early HIV testing, all HIV-infected people will eventually need ART for clinically progressive, ultimately fatal immunodeficiency. The question of when to start treatment is dominating therapeutic discussions, with frequent calls for earlier initiation of ART.[52,53] In this context, the theoretical strategy of universal voluntary HIV testing and immediate ART has some programmatic advantages. It avoids some of the major operational difficulties of ART programmes in which every patient needs to be assessed for eligibility with CD4+ cell counts and possibly viral load measurements to establish whether defined immunological or clinical thresholds, or both, for starting treatment have been achieved. Implementation of ART is simpler based solely on a positive HIV test, when most infected people are well with fairly well preserved immune systems and before many clients are lost to follow-up. The best possible drug regimen in this theoretical strategy should emphasise simplicity, low toxic effects, and ease of adherence. There will be economies of scale, allowing use of more expensive but more convenient regimens than are used in Africa at present. The supply chain for drugs could benefit from the predictable scale-up that relies on standard first-line and second-line therapy with newer regimens that are simple to use. Since increasing numbers of people living with HIV access ART, the substantial reduction in morbidity is likely to reduce the burden on the health system, freeing human resources and capacity.

Studies in Malawi and elsewhere have shown high adherence to ART, and we have no reason to believe that adherence would be reduced in special programmes,[37,54] if adequate patient support and treatment literacy is provided. Studies suggest that the development and transmission of drug-resistant strains have been restricted despite the rapid scale-up of treatment.[48] Universal voluntary HIV testing and immediate ART could raise concerns around human rights and coercion but appropriate training, engagement with the community, and supervision should keep problems to a minimum, and benefits from reduced transmission should greatly outweigh adverse results. Universal voluntary testing and treatment could reduce HIV-associated stigma and could substantially reduce the incidence of AIDS-related disease and death, including that from tuberculosis.[55]

Maximising the prevention potential of ART also has important financial implications. ART is still expensive, and concern exists over the long-term sustainability of treatment, especially if treatment is started earlier in the course of infection with less toxic but more expensive regimens. From 1996 to 2007, the yearly funding for HIV prevention, care, and treatment in low-income and middle-income countries increased 33-fold, reaching \$10 billion in 2007.[36] However, according to UNAIDS, even this amount of expenditure falls short of the estimated need. To achieve universal access—including treatment for 13·7 million people by 2010—financial resources have to nearly quadruple by 2010 from 2007, reaching \$41–58 billion by 2015.[36]

Universal voluntary HIV testing and immediate ART would entail a substantial, front-loaded investment. A full economic analysis is beyond the scope of this Article, but by changing the fundamental dynamics of the epidemic, our preliminary cost calculations suggest that there could be substantial yearly and long-term cost savings between now and 2050, by which time HIV infections could be reduced to very low and manageable amounts. Our modelling suggests that the cost of implementing the new intervention for the test-case country is much less than what UNAIDS projected for universal access to prevention, care, and treatment.

Our model suggests that massive scale-up of universal voluntary HIV testing with immediate initiation of ART could nearly stop transmission and drive HIV into an elimination phase in a high-burden setting within 1–2 years of reaching 90% of programme coverage. As with all prevention interventions for HIV, this approach should not be viewed independently of other methods of prevention. Expert evaluation and consultation with all stakeholders, including community representatives, are needed to further assess this theoretical approach and to define the role of ART in the prevention and control of HIV/AIDS.

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Conflict of interest statement
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Can antiretroviral therapy eliminate HIV transmission?

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In *The Lancet* today, Reuben Granich and colleagues (including two of us, KMDC and CFG) use mathematical modelling to assess the impact of expanded HIV testing and earlier antiretroviral therapy (ART) on HIV transmission.[1] These researchers evaluated a theoretical programme of annual universal HIV testing and immediate treatment on HIV diagnosis, irrespective of CD4+ cell count, in an HIV epidemic with southern African population dynamics. The exercise suggested that HIV transmission could be substantially reduced within a few years. Elimination of HIV transmission, defined as an incidence below one case per 1000 population per year, could be achieved within a decade, and the overall prevalence of HIV infection reduced to below 1% before the middle of the century. Compared with current practice of starting ART at a specific CD4+ count, deaths would be halved between now and 2050.

The uncontrolled number of new HIV infections, despite years of prevention effort, represents a crisis: 2·7 million incident HIV infections occurred globally in 2007.[2] By contrast, only 1 million additional individuals were placed on ART that year;[3] about 6·7 million were in danger of their lives for lack of treatment, and over 20 million other HIV-infected individuals were waiting, mostly unknowingly, for immune deficiency to progress. Universal access to HIV treatment and care is an unrealistic aspiration unless HIV transmission can be interrupted.

Behavioural interventions are essential but incompletely efficacious.[4] For biomedical interventions, only four of 24 randomised trials showed protection; three assessed male circumcision.[5] All vaccine and microbicide trials were negative. Unsurprisingly, how best to use ART for prevention has emerged as the most pressing question that faces HIV/AIDS science.

ART may be used to prevent acquisition of HIV infection when delivered as prophylaxis before or after HIV exposure, or to prevent transmission from HIV-infected persons when provided as treatment that renders them less infectious. Prophylactic approaches are challenged by the large numbers having to be treated to prevent a single infection, and ART for preventing transmission from infected persons will have the greatest impact.[6]

Biological plausibility is provided by evidence that viral load is the major risk factor for all modes of HIV transmission and that ART lowers viral load in plasma and in genital secretions.[7] Proof of concept is offered by the virtual elimination of paediatric HIV disease in high-income countries by universal voluntary HIV testing of pregnant women and appropriate provision of ART. By lowering viral load, ART reduces transmissibility as well as duration of infectiousness, the key determinants (along with rate of partner change) of the basic reproductive rate (R_0), the number of secondary infections generated by one primary case.[8] Epidemic control requires that R_0 be reduced below 1. Observational data have shown reduced HIV transmission in discordant couples after the introduction of combination ART,[9] and programmatic data supported reduction in HIV transmission at the population level.[10]

Current clinical practice, however, limits the preventive efficacy of ART because substantial HIV transmission can occur around the time of seroconversion, when viral load is highest, ample opportunity for transmission exists before reaching the CD4+ count threshold for

Reuben Granich

Sculpture by Zimbabwean Mike Munyaradzi, 2007
"Symbolizing the fragile world we live in today, the hollowed sphere signifies the destruction of our globe and the difficult issue of our time including AIDS. The leaf represents the life we are struggling to sustain."

treatment, and many HIV-infected individuals are tested late or not at all.

Dominant concerns of the approach modelled by Granich and colleagues relate to necessary protection of voluntariness and individual rights. Feasibility is challenged by: weak health systems and insufficient health personnel; choice of appropriate drug regimens; treatment adherence; drug toxicity; drug resistance and need for durable second-line and third-line regimens; the logistics, reliability, and acceptability of regularly testing a whole population for HIV infection; and behavioural risk compensation.[11] Initial costs would be high and scepticism should be expected towards large-scale top-down efforts that appear to medicalise HIV prevention.

By contrast, advantages of immediate treatment at diagnosis could include: simplified clinical management; reduction in the high mortality rates from late diagnosis;[12] control of HIV-associated tuberculosis;[13,14] and effective prevention of mother-to-child transmission of HIV, including through breastfeeding. The approach could be cost-saving over the longer term, because over time fewer individuals would be living with HIV and efforts would increasingly focus on treatment adherence and localised HIV transmission. Appropriately, Granich and colleagues emphasise that all current preventive interventions should be enhanced.

Currently, only 20% of individuals with HIV in low-income and middle-income countries know their serostatus,[3] but most infected individuals will eventually require ART if they are not to die of AIDS. Because of newly recognised, non-AIDS-defining complications of uncontrolled HIV replication,[15] as well as better treatment options,[16] the pendulum around when to start therapy is again swinging towards earlier starting.[16,17] Expanded HIV testing and immediate treatment would offer opportunity for highest quality positive prevention, a holistic approach that protects the physical, sexual, and reproductive health of individuals with HIV, and maximally reduces onward transmission.[18] Provided coercion is avoided and confidentiality and dignity maintained, individual health and societal safety should benefit through reduced HIV transmission, which would enhance human rights overall. The approach could represent a conceptual redefinition of universal access and combination prevention, reframe the debate around when to start ART, and counter prevention pessimism.

There is currently inadequate evidence for WHO to define policy and guidance about the role of ART in HIV prevention. Research is needed to assess the feasibility, safety, acceptability, impact, and cost of innovations such as the universal voluntary testing and immediate treatment approach, and broad consultation must address community, human rights, ethical, and political concerns. Research on male circumcision offers encouraging as well as cautionary experience about translation of research findings on controversial interventions into practice.[19] Discussion will also be needed on ART for prevention in populations most at risk in epidemics of varying intensity. WHO is committed to promoting consultation among countries and stakeholders about the pressing biomedical issue of ART for HIV prevention.

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Treating our way out of the HIV pandemic: could we, would we, should we?

HIV prevention is easy in theory—the practice is hard. In models, HIV can be eliminated if risk behaviours or viral transmissibility are reduced substantially. Unfortunately, in many places, achievement of these reductions has not been possible and HIV incidence has remained high. In *The Lancet* today, Reuben Granich and colleagues use mathematical models to show that annual screening of most adults for HIV, with immediate commencement of antiretroviral therapy (ART) for all those infected, would dramatically reduce HIV incidence.[1] This strategy would be a bold move away from the current approach of treatment on the basis of clinical need, in which the hoped-for synergies between treatment and prevention will remain limited because testing and counselling focus on individuals who have probably already transmitted infection.[2]

The "treat early, treat hard" approach of the early era of triple ART fell out of favour due to drug toxicities, concerns over the evolution of drug resistance, and potential limits to the duration of treatment efficacy in the patient. Therefore, when Velasco-Hernandez and colleagues first showed in simple models that ART could eliminate HIV infection, their findings appeared extremely unrealistic.[3] Since then, clinical opinions have changed with improved regimens and evidence of lower death rates if ART is started before CD4+ cell counts fall below 350 per µL, which makes earlier treatment more acceptable.[4]

When early treatment is considered as a prevention tool, success will require substantial resources and depend on a remarkable degree of acceptance and cooperation across populations. If we could eliminate HIV this way would we, given the will needed, and should we, given the conflict between utilitarianism and individualism[5] inherent in this strategy?

Granich and colleagues' model findings appear robust and the critical mathematics is straightforward. At the start of an epidemic, the number of subsequent infections generated by each infection (the basic reproductive number, R_0) determines how difficult it will be to control the infection.[6] Reductions in variables that contribute to R_0, such as the average transmission probability, will reduce the spread of infection. There is a simple rule that, for a given value of R_0, an equivalent fold reduction, reducing R_0 to its tipping point of one, will eliminate the infection. Secondary infections over the course of infection and when individuals need to be treated are shown in the figure. To eliminate infection, the individual must be treated and become non-infectious at the point when the cumulative number of infections passes through one. To allow for early high infectiousness and potentially high R_0 values, Granich's model represents a very intensive screening programme. Such a programme and additional interventions in those with highest risk would probably be required if elimination was truly the goal. However, real elimination (ie, an incidence of zero in a large geographic area[7]) is unlikely. There will be some people whose risk is sufficient to maintain pockets of transmission within a very small fraction of the population. Probably, some individuals would not be screened and treated, and

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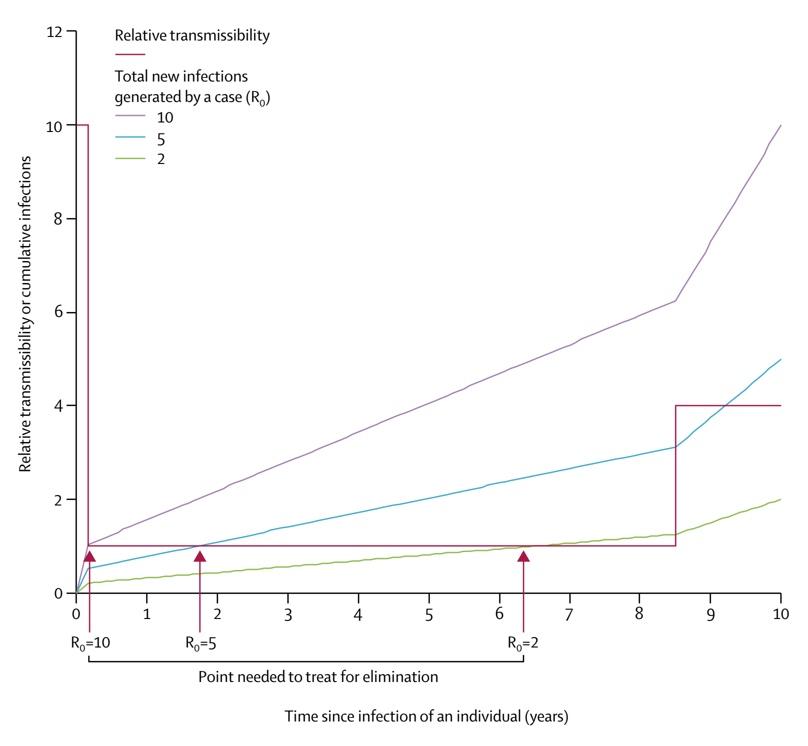

***Figure:* Cumulative number of new HIV infections transmitted per infected individual**
As function of time since infection for scenarios in which each individual generates two, five, and ten new infections. Infectiousness is assumed to be ten-fold higher in first 2 months than in long asymptomatic period, and four-fold higher in final 1·5 years of average infection of 10 years.

if these individuals have high risks of HIV acquisition and transmission, they will maintain infection within the population. Nonetheless, dramatic reductions in incidence are likely to be achieved.

Granich and colleagues' suggested strategy would be extremely radical, with medical intervention for public-health benefits rather than individual patient's benefits; more so, if the costs of toxicity and inconvenience for the individual outweigh the clinical benefits. Because screening and treatment would be for the public good, resources would have to come from the public purse. Current studies of uptake of and adherence to ART represent the case when mostly sick people are motivated to seek care. The strategy would rely on people, who feel well, regularly seeking care and adhering to ART. The prevention impact of the treatment depends on those who receive ART having a much reduced transmissibility, which seems reasonable, but there would be concern over other sexually transmitted infections, poor adherence, and treatment failure all reconstituting infectiousness, or the evolution and spread of resistant virus undermining treatment. There would also be questions about the epidemic contexts in which such a strategy should be used, the required frequency of testing, and how that frequency would change as prevalence falls.

The suggested strategy would reflect public health at its best and its worst. At its best, the strategy would prevent morbidity and mortality for the population, both through better treatment of the individual and reduced spread of HIV. At its worst, the strategy will involve over-testing, over-treatment, side-effects, resistance, and potentially reduced autonomy of the individual in their choices of care. The individual might gain no personal benefit from testing and early treatment, but they would benefit from protecting partners—and who could object to that, unless they were recklessly exposing others to infection? It is easy to see how enforced testing and treatment for the good of society would follow from such an argument. Partial success would lead to infection becoming concentrated in those with a high risk, with an increased danger of stigma and coercion. The history of the control of sexually transmitted infections documents several examples of compulsory screening and treatment of stigmatised populations,[8] and there is a danger of a well-meaning paternalistic medical model following such a route.

Capacity—financial, physical, and human—as well as effectiveness and cost-effectiveness are important. The relative power of curative versus preventive medicine and the priority of health enforcement would be crucial in implementation. Challenges will rightly come from those concerned about individuals' rights and patients' autonomy, as well as from those who moralistically fear an "easy" solution to HIV rather than behavioural change.

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Supporting on-line information

Universal HIV Testing with Immediate Antiretroviral Therapy as a Strategy for Eliminating HIV Transmission

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Initial distribution of $CD4^+$ cell counts

The initial distribution of $CD4^+$ cell counts, shown in Figure 1, is taken from a random sample of young adult men in Orange Farm, Gauteng, South Africa.[1] The fitted curve is log-normal with a median at 1116/μL and a standard deviation of 0.303.[1]

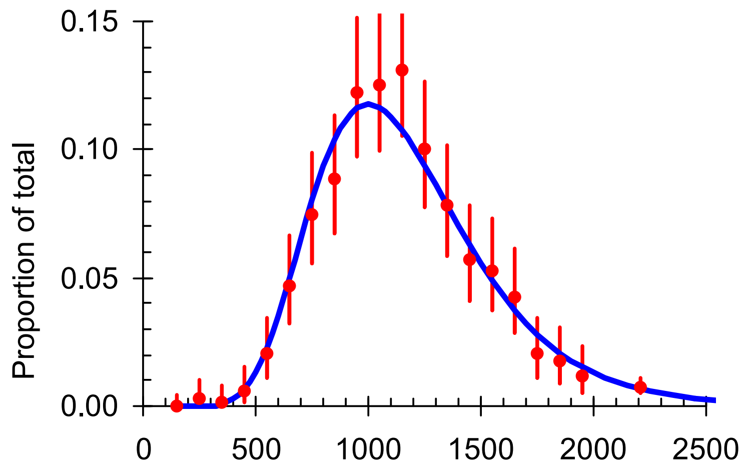

Figure 1. The observed distribution of $CD4^+$ cells in HIV-negative men in South Africa.[1]

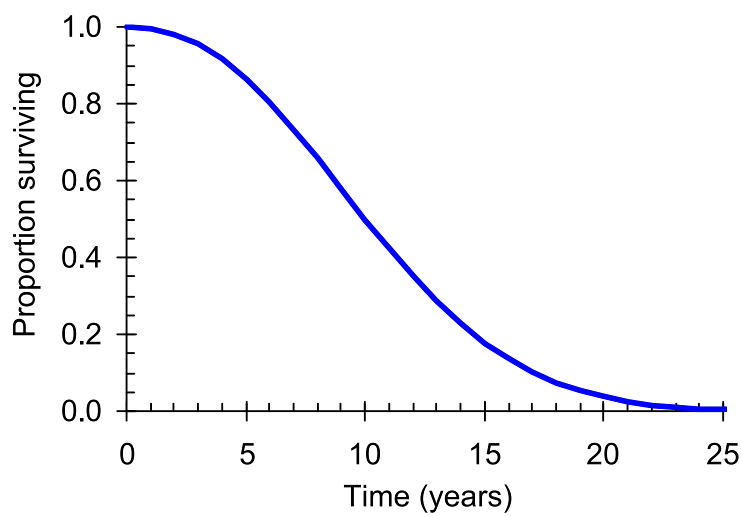

Figure 2. The Weibull survival curve used in the model based on data from the CASCADE study.[2]

Survival distribution

The models are not age structured. We therefore assume that survival follows a Weibull distribution[3] with a median of 11.0 years and a shape parameter of 2.25[4] as illustrated in Figure 2.

Stochastic model for estimating R_0

Variation in infectivity with time

For a person with a given survival time, we allow the infectivity to vary with time as illustrated in Figure 3. We consider three phases which we will call acute, chronic and final. The infectivity can be varied in each of three phases as indicated by the letters *a* to *c* in Figure 3, and the duration of the acute and final phases can be varied as indicated by the letters *d* and *e* in Figure 3.

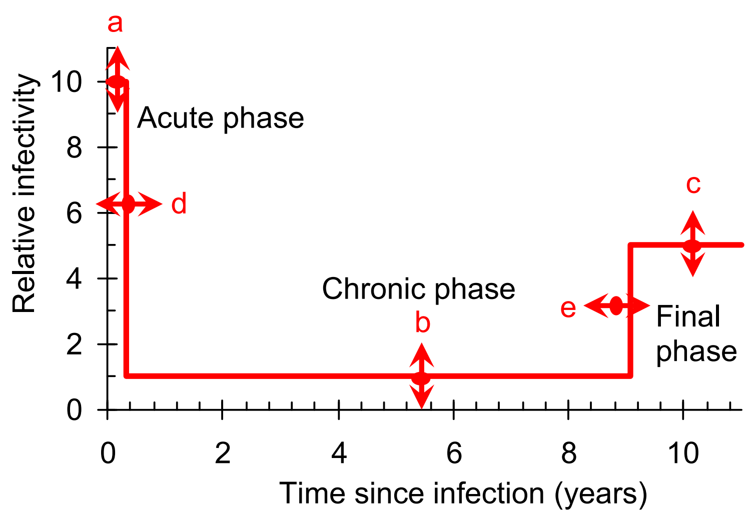

Figure 3. Schematic diagram of the change in infectivity with time in a person who survives for ten years. There is an initial acute phase with high infectivity, a chronic phase when the infectivity is lower and a final phase when the infectivity increases again. The infectivity during each of the three phases (a–c) can be varied, as can the duration of the acute and final phases (d and e).

The parameter values in Figure 3 are not known precisely. In a study in Uganda the HIV transmission probability per coital act was measured directly between discordant couples[5,6] and the transmission probability during the first five months after infection and the last three years before death was compared to the transmission probability during the chronic phase. Two other studies used the same data but obtained slightly different rates.[7,8] In a study in the USA viral load in plasma and in semen was measured as a function of time since infection with HIV and the transmission rates during the acute and final phases were compared to the rates during the chronic phase among men who have sex with men.[9] In a study in San Francisco, USA, the transmission probabilities from men with known dates of sero-conversion to their female partners were compared 20 and 54 days after sero-conversion.[10] The results of these studies are given in Figure 4. For our base-line calculations we assume that the acute phase lasts for two months during which time the probability of transmission is ten times greater than during the chronic phase and that the final phase lasts for 10% of the survival time (1.1 years when survival is eleven years) and during this time the probability of transmission is five times greater than it is during the chronic phase.

We adjust the chronic phase infectivity to give the appropriate value of R_0, the mean number of secondary infections that arise from one primary infection, per year. The infectivity in the acute and chronic phases is then calculated as described above.

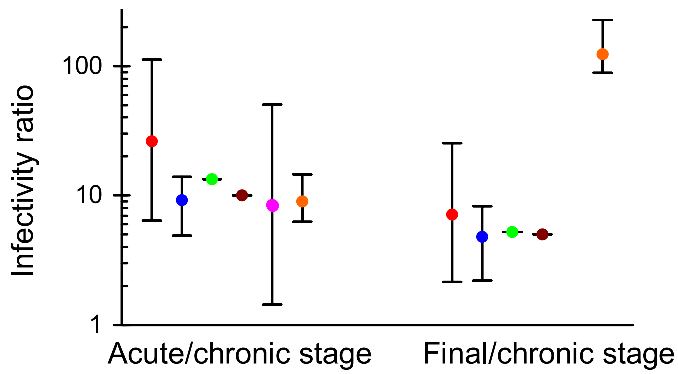

Figure 4. Estimates of the ratio of the infectivity per sexual contact in the acute and final phases to the chronic phase. Red, blue and green points are three different analyses using data from Rakai, Uganda,[6-8] the brown points give the values used in this paper, the pink point is for transmission to the female partners of men with known dates of sero-conversion in the United States[10] and the orange points are for receptive anal intercourse in a study of men in San Francisco.[9]

Partners and concurrency

For the baseline calculations the number of concurrent partners is four and the mean duration of partnerships is six months so that people have an average of eight partners per year. At each time step any one of the current partners may be changed with a probability calculated to give the preset mean duration of a partnership.

Model structure

The model runs as follows. Let $P(C)$ be the log-normal distribution of initial CD4$^+$ cell counts; $P(S)$ be the Weibull survival distribution; $p(a)$, $p(c)$ and $p(f)$ be the probabilities, per unit time step, that the index case infects any one partner during the acute, chronic or final phase of infection, respectively; $p(n)$ and $p(h)$ be the probabilities, per unit time step, that the index case changes any one partner or is tested for HIV, respectively.

1 Choose a starting CD4 cell count randomly from $P(C)$.
2 Choose a survival time randomly from $P(S)$.
3 During each step:
   1.3 Check if the person is in the acute, chronic or final phase of infection;
   2.3 Infect another person with probability $p(a)$, $p(c)$ or $p(f)$ depending on the phase of infection;

3.3 Change one or more partners, each with probability $p(n)$;

4.3 Test the person for HIV with probability $p(h)$;

3.4.1 If the person is, or has previously been, tested (and is, by definition, HIV-positive) check the persons CD4 cell count.

3.4.1.1 If it is below the CD4 threshold → Stop

3.4.1.2 If it is above the CD4 threshold increase time, calculate a new CD4 count → 3.

3.4.2 If the person has not been tested check if the CD4 cell count is greater than zero.

3.4.2.1 If the CD4 cell count is less than or equal to zero → Stop

3.4.2.2 If the CD4 count is greater than zero increase time, calculate new CD4 count → 3.

Illustration of the model

The baseline parameter values are given in Table 1. The model, which runs in Excel using Visual Basic, is available from the senior author on request. The model runs with a time step of one month. At each time step other people may be infected, any of the partners may be changed, and the index case may be tested for HIV. Each of these Poisson processes happens with a fixed probability chosen so that the mean rate of infection, duration of partnerships or rate of testing match the values defined by the parameters in Table 1 and the initial $CD4^+$ cell count and the survival are chosen from the distributions specified. Random numbers are generated using a Mersenne-Twistor.[11,12]

Model output

The output of the model is illustrated in Figure 5 to Figure 7. Figure 5 shows the distribution of the number of secondary cases which ranges from 0 to 25 with a mean of 7.4. Figure 6 shows the distribution of the time $\overline{\tau}$ between the primary infection and each secondary infection. The mean value of $\overline{\tau}$ is 7.54 years but note that many more people are infected during the acute phase in months 1 and 2 than in any other month although the acute phase infections are still only 8.9% of all infections. The chronic phase infectivity was chosen to give a doubling time of R_0 of 7.4 and a mean time to each secondary infection of 7.54 years so that the epidemic doubling time is $\tau_0 \approx \overline{\tau}/(R_0 - 1) =$ 1.2 years, close to the observed value for South Africa[13] (see Figure 4B in the main text). Figure 7 provides a check that, if ART is not provided, the stochastic model produces the preset Weibull survival distribution.

Table 1. Parameter values for the log-normal distribution of the initial CD4 cell count (Figure 1); the Weibull survival (Figure 2); the chronic phase is defined in terms of the number of people who are infected each year (on average); the acute phase which is a step function with a defined mean duration and the infectivity set relative to that in the chronic phase; the final phase which is a step function with a preset infectivity and duration set to a fixed proportion of the survival in a particular run; the number of concurrent partners in a given simulation is fixed as is the mean duration of each partnership. Infection is a Poisson process as is the probability of changing any partner with rates set to match the preset mean values. The CD4 count at which ART begins, if the person has been tested is fixed. The last row sets the number of Monte Carlo repeats.

Parameter definition		Value	Dimensions
Initial CD4	Mu	6.98	Log normal location parameter
	Sigma	0.34	Log-normal shape parameter
Survival	Median	10.0	Years
	Shape	2.25	Dimensionless
Acute phase	Duration	2	Integer months
	Infectivity	4.50	Number/year
Chronic phase	Infectivity	0.45	Number/year
Final phase	Infectivity	2.25	Number/year
	Duration	0.15	Proportion of expected life
Concurrency	Number	4	Number
Partnership	Duration	6	Months
Start ART	CD4	0	/micro-litre
Proportion tested	Rate	0	/year
Monte Carlo	Repeats	10,000	Number

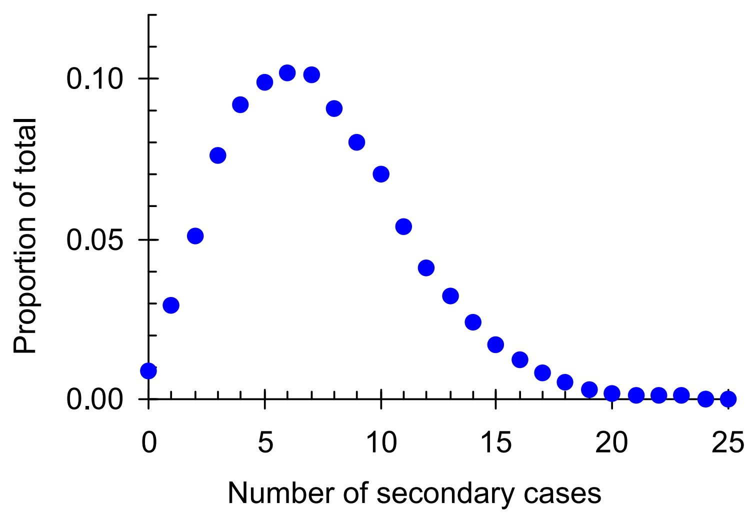

Figure 5. The frequency distribution of the number of secondary cases generated by one primary case. The mean number of secondary cases is 7.4.

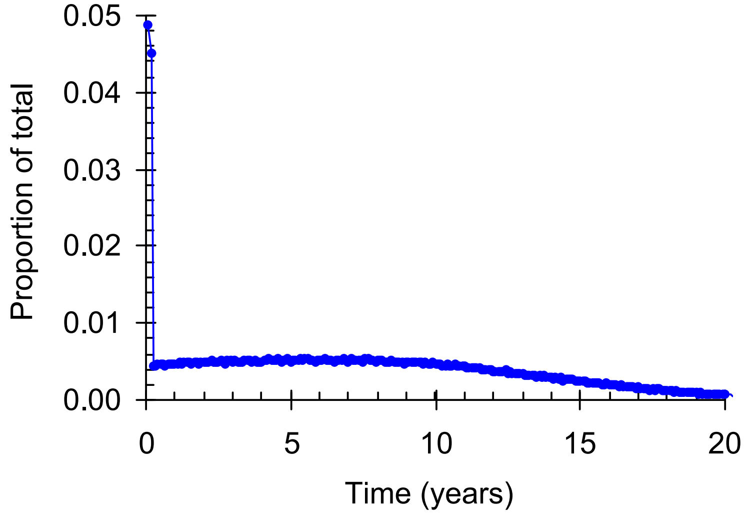

Figure 6. The frequency distribution of the time between the infection of the index case and the infection of each secondary case. Transmission is highest during the first two months but only contributes 8.9% of all secondary infections.

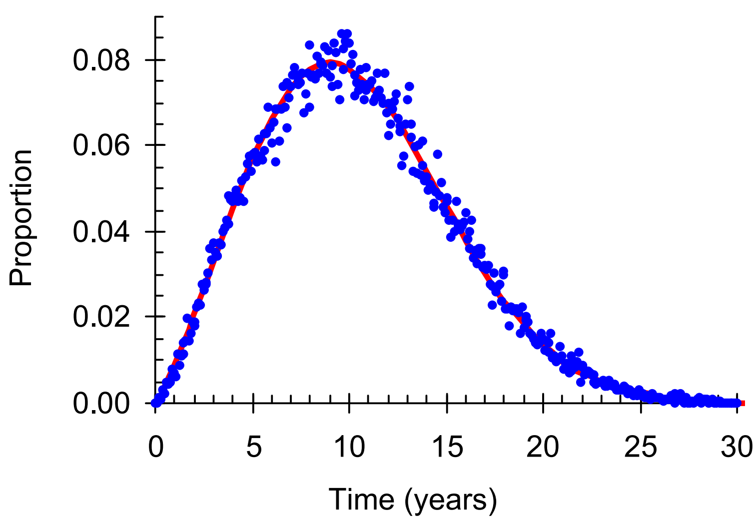

Figure 7. The frequency distribution of the time to death from the simulation (blue dots) compared to the distribution obtained from the Weibull curve if no one is started on ART, as confirmation of the model.

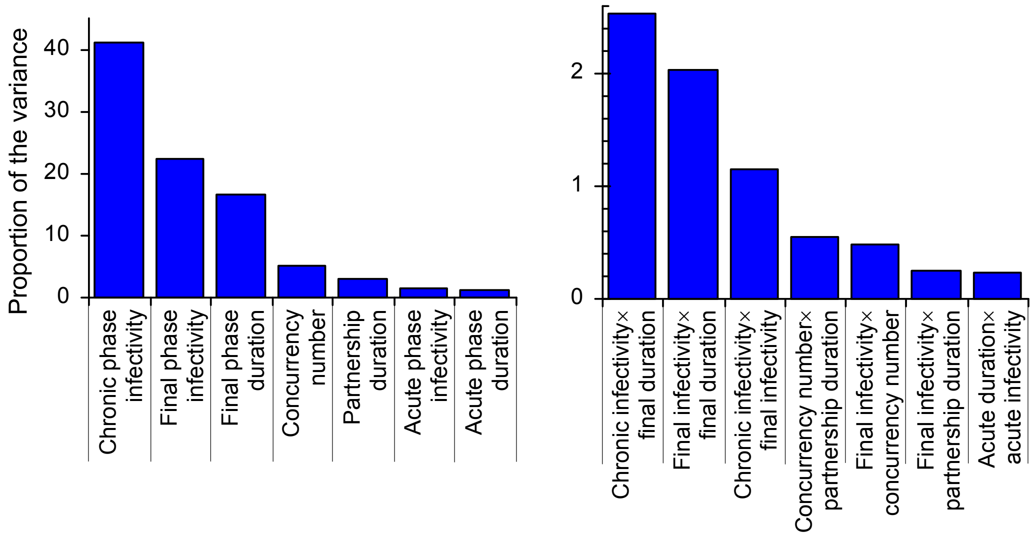

Figure 8. Proportion of the variance in the average value of R_0 explained by the variation in each parameter. Considering the individual terms, 80% of the variance is explained by the chronic and final phase parameters and 11% by the concurrency number, the partnership duration and the acute phase parameters. Considering the interaction terms 5.7% of the variance is explained by the chronic and final phase parameters and 1.5% by the concurrency number, the partnership duration and the acute phase parameters.

Sensitivity analysis

In order to explore the sensitivity of the estimated values of R_0 to the various parameters in the model we ran one thousand simulations with the base line set of parameters from which we calculated R_0 and the intrinsic doubling time τ_0. We repeated this ten thousand times and on each repeat each parameter was kept the same or multiplied or divided by 2 with equal probability. The ten thousand results were then imported into *Stata* and a multivariate analysis of variance was done with $\ln(R_0)$ as the dependent variable and the full set of parameters as the independent variables (Figure 7). (A similar analysis was done for $\ln(\tau_0)$; results not shown).

We examined the role of the high levels of infectivity during the acute phase when people have several concurrent partners because the dynamic model, with which we explore trends in prevalence, incidence and mortality as we approach the elimination phase of HIV, does not include a separate acute phase. The analysis-of-variance (Figure 8) shows that the terms in the model involving the chronic phase and the final phase account for 80% of the variance (86% including their interactions). The terms involving concurrency, partnership duration and the acute phase account for 11% of the variance (13% including their interactions). The model output is therefore relatively insensitive to the acute phase parameters as compared to the chronic phase parameters for similar proportional changes in the parameters.

Table 2. The coefficients of the regression of $\ln(R_0)$ and $\ln(\tau_0)$ against the various independent variables. The table gives the constant term, the regression coefficients and the change in R_0 or τ_0 if the parameter is doubled or halved from the based line values given in Table 1. The standard deviations of the estimates of each co-efficient (arising from the Monte Carlo simulation) are all less than 0.002. The base-line value of R_0 is $e^{2.058} = 7.84$ and of τ_0 is $e^{0.073} = 1.08$ yrs.

Variable	Coeff. (R_0)	ΔR_0 (%)	Coeff. (τ_0)	$\Delta\tau_0$ (%)
Constant	2.058	•	0.073	•
Chronic phase infectivity	0.303	35	–0.420	–34
Final phase infectivity	0.226	25	–0.160	–15
Final phase duration	0.192	21	–0.152	–14
Concurrency number	0.108	11	–0.134	–10
Duration of partnerships	–0.082	–8	–0.117	–8
Acute phase infectivity	0.063	6	–0.109	–13
Acute phase duration	0.054	6	0.073	–11

Table 2 shows the proportional change in R_0 when each of the parameters is increased by a factor of two. The value of R_0 is about two to three times less sensitive to changes in the parameters defining the number and duration of partnerships and about four to five times less sensitive to changes in the parameters defining the acute phase than it is to the parameters defining the chronic and final stages of infection. If we were to eliminate the acute phase from the model, by setting the infectivity during this phase and during the chronic phase to be the same, we would have to reduce the acute phase infectivity by a factor of ten. To maintain the value of R_0, we would need to balance this ten-fold reduction in the relative infectivity of the acute phase by the increasing chronic-phase infectivity by a factor of 1.6. These results show that the overall level of transmission is dominated by the chronic and, to a lesser extent, the final phase, rather than by the acute phase. In a subsequent publication we intend to explore in more detail the affect of each of the parameters in the model to transmission.

We carried out a similar analysis to explore the sensitivity of the doubling time to each of the parameters. The results, also given in Table 2, are similar to those for R_0 except that the result is less sensitive to the final phase infectivity and more sensitive to the acute phase infectivity, as one would expect given the long time scale over which the latter is effective and the short time scale over which the former is effective.

Summary

The stochastic model suggests that the results depend mainly on the overall level of transmission rather than transmission during the acute phase or the number of concurrent partners. In essence, if each HIV-positive person infects one person every one to two years, on average, then even allowing for a ten-fold increase in infectivity during the acute phase they will only infect one person every one to two months during the acute phase no matter how many concurrent partners they have. Furthermore, if the acute phase was mainly responsible for driving the epidemic then the doubling time would be of the order of months, not years.

Dynamical model to assess impact of interventions

The dynamical model is illustrated and described in Figure 3 in the main text. Here we discuss the assumptions underlying the structure of the model and give details of the estimation of some of the parameters.

Survival on and off ART

The model has four infected classes so that the survival is a gamma-function, $\Gamma(\tau, 4)$. A Weibull function with shape parameter 2.25 and mean survival equal to 11 years has a median survival of 10.6 years. The best fit gamma-function has τ = 2.8 years so that people move between classes at a rate $\omega = 1/\tau$ = 0.36/year.[14] If people are tested and found to be positive they move to the equivalent ART stage, at a rate τ, and continue to progress from there (Figure 2 in the main text). If people are non-compliant they move back from the ART boxes to the equivalent non-ART boxes at a rate ϕ.

The survival on ART after being found to be HIV-positive in each of the four compartments is important to the model but difficult to determine precisely. A recent paper[15] suggests that survival on HAART is about 6 to 8 years if people start when they are symptomatic and their CD4$^+$ cell count is less than 100/µL, and about 15 to 18 years if people start HAART when they are asymptomatic and their CD4$^+$ cell count is > 350/µL. Here we let $\sigma = 2\tau$. Since the mean survival time without ART is eleven years the mean duration in each of the four compartments for infected people is 2.8 years. Since the mean CD4$^+$ cell count in Figure 1 is 1179/µL and there is an initial decline of 25% immediately after seroconversion,[16-19] the mean CD4$^+$ cell count in each of the compartments is 774/µL, 553/µL, 332/µL and 111/µL. When HIV-positive people are started on treatment they move from one of the four compartments I_i to the corresponding compartment A_i. The average life-expectancies of people who are detected in states I_1 to I_4 and who start ART immediately, are therefore 21, 18, 15 and 12 years, respectively.

Adherence to ART

The conditions for eliminating HIV are sensitive to the level of adherence. We stop transmission by putting everyone who is HIV-positive on ART so that there are a very large number of people living on ART and if a small proportion of them stop taking their drugs they will contribute a significant number of infectious cases to the population.

Field data under programme conditions are available from Malawi (A.J. Harries, person communication). Here we use the results of the Malawi cohort analyses that have been reported for each quarter for the last two years. In these analyses reports were made 6, 12, 18, 24 and 36 months after people started ART (although not all of these time periods were reported in each report). In each quarter we start with the number of people who started treatment the appropriate number of months earlier which ranges from three thousand to thirteen thousand, the total number of reports in all the cohorts combined being 163 thousand. The possible outcomes were: Alive and on ART; Transferred out; Dead; Lost to follow up; and Stopped treatment.

In the quarterly reports for Malawi 19% of those who started treatment were 'transferred out' and, in the reports, are presumed to be still alive. Here, we remove them from the numerator and denominator which is equivalent to assuming, more conservatively, that their outcomes are the same as the outcomes for those who remain in the cohort. A substantial proportion of people die, 16% of all those that started treatment, in part because most people in Malawi start ART with fairly advanced infections. Those who die reduce the total number of people who are on ART so that leaving them out of the model makes the model more conservative. However, in all of the cohorts 11% of people were lost to follow up and 0.7% stopped treatment and we need to decide what proportion of people who were lost to follow up also stopped treatment; they may have died or they may have started treatment elsewhere.

Table 3. Cohort analyses from the ART programme in Malawi. For each quarter the table gives the number of months before that quarter when people started ART. It then gives the number of people who started treatment and the proportion of these that were lost to follow-up the appropriate number of months later.

	2007 Q4		2007 Q3		2007 Q2		2007 Q1		2006 Q4		2006 Q3		2006 Q2	
Mnth.	Start. (k)	Lost (%)	Start. (k)	Lost (%)	Start. (k)	Lost (%)	Start. (k)	Lost (%)	Start. (k)	Lost (%)	Start. (k)	Lost (%)	Start. (k)	Lost (%)
6					11.8	8.6	11.0	8.4	9.8	7.3	8.3	7.9	7.1	9.9
12	11.2	11.6	11.0	10.0	9.7	8.6	8.0	10.9	7.1	13.0	7.1	9.2	4.6	8.6
18					6.6	11.5	6.7	10.7	4.7	10.6	4.0	10.1	2.5	11.6
24	6.8	14.1	6.6	10.9	4.5	10.9	3.9	13.1	2.6	13.7				
36	1.9	16.6	2.0	13.4										

The data in Table 3 are plotted in Figure 9. The fitted line is an off-set exponential of the form:

$$L(t) = \alpha + \left(\beta - \alpha\right)\left(1 - \mathrm{e}^{-\gamma t}\right) \qquad 1$$

It is difficult to be certain about the rate of loss in the first six months but the fit in Figure 9 suggests that the initial loss to follow-up is about 7%. After the initial drop-out the loss to follow-up is 2.9% per year falling to 2.2% per year after three years. A recent study in Malawi[20] traced 737 who had transferred out and found that 92% of them had transferred in to another programme with a median delay of 1.3 months. If these numbers apply broadly then the loss to follow up, after the first six months, is about 0.2% per year. To this we need to add the 0.7% per year who are known to have stopped treatment. The initial drop-out rate from ART, excluding those that die, may be as high as 8% and the subsequent drop-out rate is likely to be between 1% and 3% per year. Here we assume a long-term drop out rate (excluding deaths) of 1.5% per year.

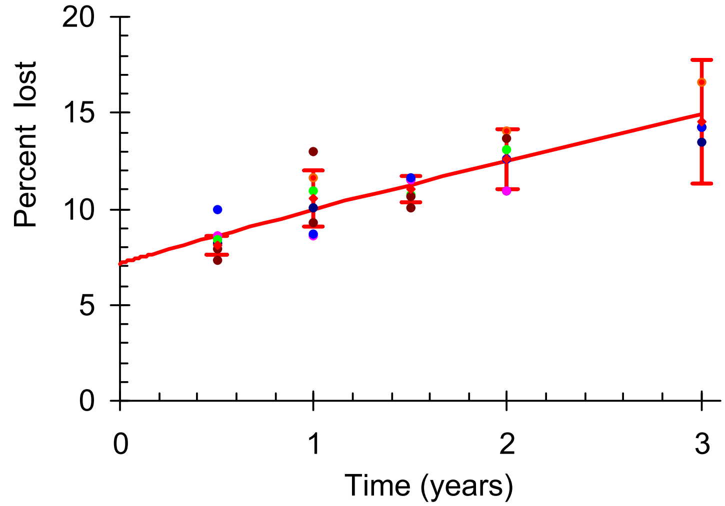

Figure 9. Cumulative loss to follow-up for the Malawi cohort (Anthony Harries, personal communication). Dots are data from particular cohorts. Orange diamonds give average values with 95% error bars. The solid line is an off-set exponential (Equation 1) with $\alpha = 7.2$, $\beta = 43.1$, $\gamma = 8.1$/yr.

Transmission on ART

We need to consider the residual transmission when people are on ART. We assume people are either completely compliant with their treatment or, if they fail, they return to the untreated state. While people who are fully compliant with respect to their drugs are unlikely to infect others, there may be some residual transmission. A recent study of the reduction in viral load[21] suggests that there is a two-log decline in viral load over the first twenty days after starting ART treatment and a further two- to four-log decline over the next 150 days after starting treatment. The reduction seems to be the same for those starting treatment relatively early, with a median $CD4^+$ cell count of 444/μL, as it is for those starting late, with a median $CD4^+$ cell count of 13/μL. However, adherence is important and there is viral rebound among those who do not fully comply with their treatment.

Data on the relationship between viral load and transmission are limited. However, a study in Uganda suggests that transmission declines as the viral load to the power of 0.4[22] so that that reducing viral load by between 10^6 and 10^4 times would reduce transmission to between 0.4% and 2.5% of its value when people are not on ART. Direct data are also limited and the few published studies generally involve people in the late stages of infection or on mono-therapy.[23,24] In a study of discordant couples in Uganda[25] transmission was reduced to 2% (0.05%–11%) of it pre-ART level. Here we assume for our base-line calculations that ART reduces transmission to 1% of the value when people are not on treatment. The extent of residual transmission among people who are on ART is of great importance and this is subject to a sensitivity analysis in Figure 10g (below).

Rate of spread of programme coverage

We assume that coverage increases logistically as shown in Figure 10. In 2007 UNAIDS estimates were that about 5% of all those living with HIV were on ART (250k out of 5.3M). We also assume that coverage reaches 50% in 2011 and that it reaches 95% in 2016.

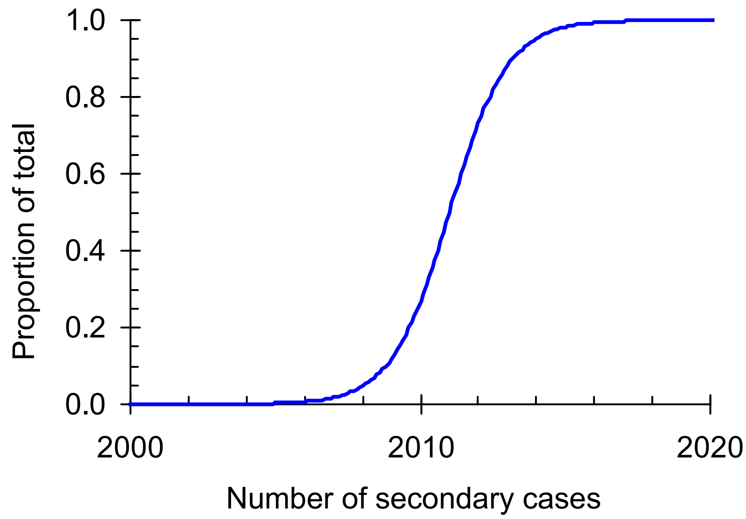

Figure 10. The coverage or extent of the intervention as a function of time. We assume that at full coverage 90% of the population is reached.

Comparison with other control methods

We compare the impact of the proposed strategy with the impact of other methods of reducing transmission which might include partner reduction, condom use, treatment of curable sexually transmitted infections, male circumcision and so on.

Figure 11 shows the impact on the epidemic if transmission is reduced by 20%, 40%, 80% and 95%. If transmission is reduced by 40% over the next ten years this will only reduce the incidence in 2030 to 1.2% per year (Figure 11B). To achieve a reduction equivalent to that of the proposed strategy (Figure 4A in the main text) would need a reduction in transmission of 95% over the next ten years (Figure 11D).

Sensitivity analysis

It is important to examine the sensitivity of the results to the various parameters in the model. Since we are concerned mainly with the conditions under which we can eliminate transmission we focus the sensitivity analysis on the effect that varying the model parameter has on the incidence, prevalence and mortality among people not on ART in 2030.

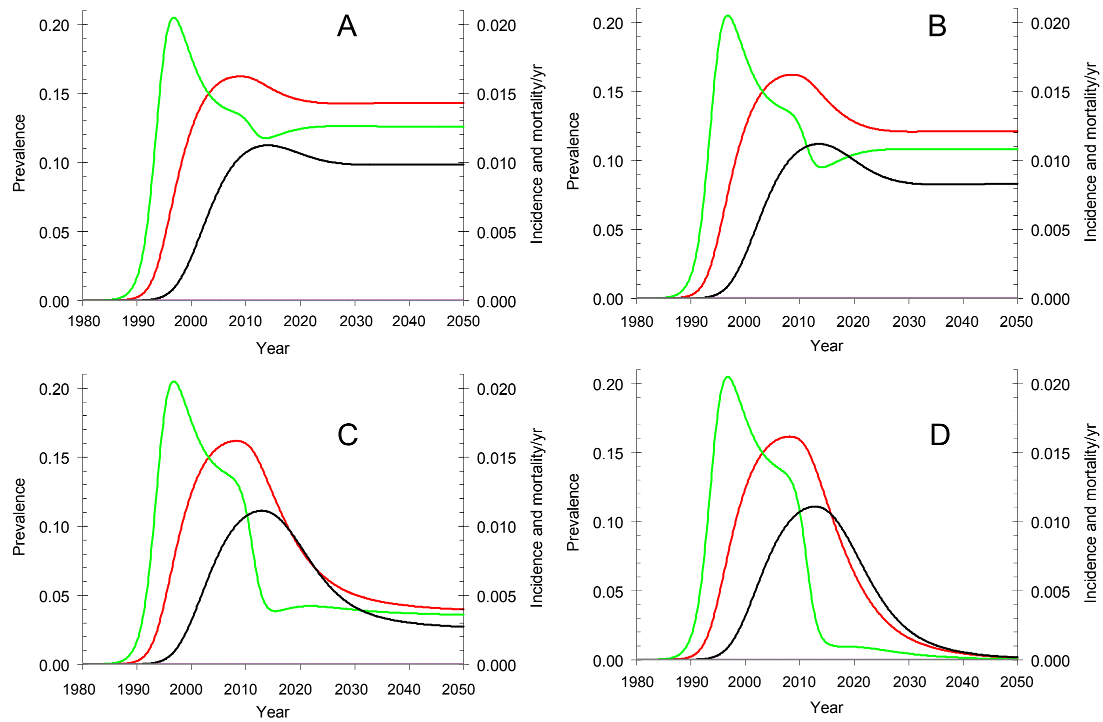

Figure 11. The impact of interventions that reduce transmission on the prevalence (red), incidence (green) and mortality (black) of HIV in South Africa. The reduction in transmission follows the curve shown in Figure 10 and the final reduction is A 20%, B 40%, C 80% and D 95%. The 95% reduction in transmission shown in D is the reduction needed to match the impact of the proposed strategy on incidence in 2030 as shown in Figure 4A of the main text.

Figure 12 shows a univariate sensitivity analysis in which the parameter values are varied about their default values given in Table 4. We plot the prevalence, incidence and mortality among those that are not on ART in 2030 against each parameter. Among adults, at the default values, the prevalence is 0.10%, the incidence is 0.042% per year, and the mortality is 0.002% per year as indicated by the large dots in Figure 12.

Figure 12A shows that the impact is insensitive to the rate at which the intervention is implemented bearing in mind that we assume that the intervention reaches halfway at the same year so that if it takes longer it also starts earlier. Starting the intervention later would simply delay the impact.

Figure 12B shows that if we also ensure that people with low $CD4^+$ cell counts are identified and started on ART, then we reduce mortality by a further factor of ten but have little additional impact on incidence or prevalence, as expected on the basis of the Monte Carlo model.

Figure 12C shows that the results are very sensitive to the average testing interval. To keep the incidence of new infection below 0.1% per year we need to ensure that people are tested at least once a year, on average.

Figure 12D shows that the results are insensitive to the proportion refusing treatment essentially because they do not start ART and do not contribute to the build up of people on ART whose breakdown leads to the residual transmission in 2030. Once the elimination phase has been reached there are few new incident cases and the refusal rate does not have a major impact on transmission.

Table 4. Base-line parameter values. The demographic, transmission, progression, drop-out and testing rates refer to the parameters in Figure 3 of the main text. The intervention parameters define the logistic curve which gives the rate at which the interventions are rolled out. 'Testing refusal' gives the proportion of people who refuse testing so that the actual testing rate is $\tau(1-\xi)$. The '*CD4 failure rate*' gives the proportion of people who, having been tested at a low CD4 cell count, fail treatment and die. The parameter for '*other interventions*' gives the reduction in transmission due to condom use, behaviour change or any other complementary intervention. We assume that these are rolled out following the same logistic curve defined in the table.

Parameter		Value	Units
Birth rate	β	0.029	/yr
Background mortality rate	μ	0.018	/yr
Transmission parameter	Initial value λ	0.767	/yr
	Location α	0.055	0
	Shape n	0.996	0
	Relative transmission on ART ε	0.010	
Progression	Untreated ρ	0.303	/yr
	Treated σ	0.165	/yr
Intervention	Timing t	2011	yr
	Rate r	1.00	/yr
	Maximum m	0.90	
Testing	Rate τ	1.00	/yr
	Refusal ξ	0.08	
Drop-out	ϕ	0.015	/yr
CD4	On/Off	Off	
	Failure rate η	0.10	/yr
Other interventions	Reduction in transmission γ	0.40	

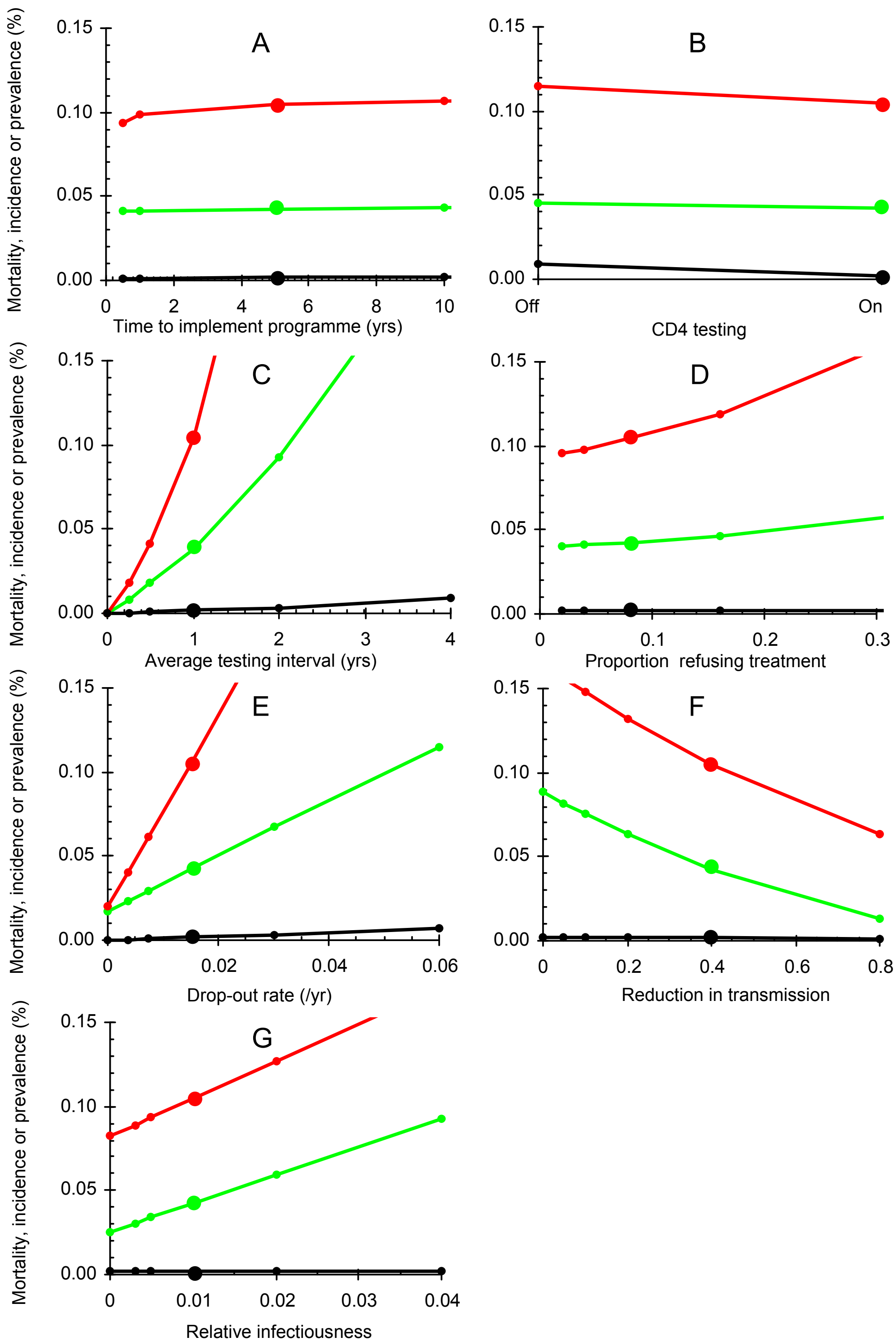

Figure 12. Sensitivity analysis. Red: prevalence; green incidence; black mortality, in 2030. A The time it takes to implement the programme; B The effect of introducing testing at low CD4 cell counts, C The rate of testing; D The proportion of people that refuse treatment; E The rate of drop-out from the ART programme (excluding deaths); F The reduction in transmission if a complementary intervention is introduced at the same time. Large dots give the values for the base-line parameter values.

Figure 12E shows that the results are sensitive to the rate at which people drop-out of the ART programme and return to an infectious state. To ensure that the incidence is less than 0.1% per year in 2030 fewer than 5% of people should drop-out of the ART programme each year. The reason for this is that when the elimination phase has been reached there are very many HIV-positive people living on ART and if only a small percentage of them drop-out this has a big impact on transmission.

Figure 12F shows that while other complementary interventions have an impact on prevalence and incidence they have much less impact on mortality as this depends on people who are already infected progressing to death over a period of ten years.

Finally, Figure 12G shows that incidence and prevalence are sensitive to the relative infectiousness of people on ART compared to those not on ART. To keep the incidence of new infections below 0.1% per year the infectiousness of those on ART should be less than 5% of that of those not on ART.

Summary

The key parameters are the testing rate, the drop out rate and the relative infectiousness of people on and off ART. If the intention is to ensure that there is less than one new infection per thousand people per year in the elimination phase, then we need to test people at least once a year on average, ensure that the drop-out rate from ART is less than 5% per year and ensure that the viral-load suppression is sufficient to the infectiousness of those on ART to less than 5% of that of people not on ART.

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AIDS RESEARCH

Treat Everyone Now? A 'Radical' Model to Stop HIV's Spread

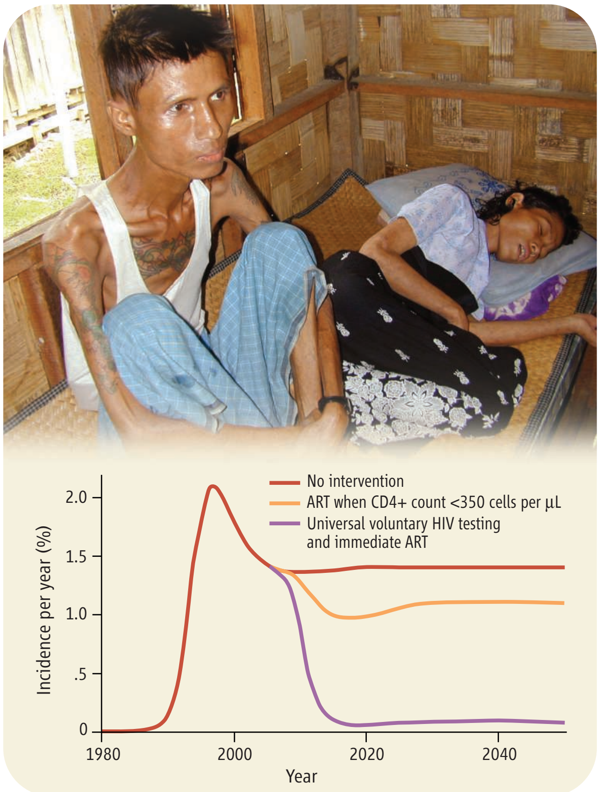
Confronted with the lackluster success of HIV prevention efforts, researchers are increasingly pushing treatment itself as a way to slow the spread of the AIDS epidemic. The reasoning: Antiretroviral (ARV) drugs lower the amount of HIV in infected people, likely making them less able to transmit the virus. Now the World Health Organization (WHO) has published a provocative model that explores the possibility of "eliminating" the HIV epidemic by annually testing everyone on a voluntary basis and treating all infected people, regardless of their clinical status. WHO HIV/AIDS Director Kevin De Cock, who co-authored the study that appeared online 26 November in *The Lancet*, says, "What we hope to do is stimulate discussion." And that they have.

Testing and treatment on that scale would far surpass anything done today, but the payoffs could be huge, the authors say. According to their model, each untreated infected person infects seven others before dying. The study compares the impact of thwarting new infections when people start treatment at various stages of immune decline, and it assumes that treatment reduces a person's infectivity by 99%. Most poor countries can afford to treat only the people most in need, which is defined as anyone who has fewer than 200 CD4+ white blood cells, the critical immune warriors that HIV targets and destroys.

As an example, the study focuses on South Africa, which has an adult prevalence of 17% and offers treatment at the higher cutoff, common in wealthy countries, of 350 CD4+s. At that point, the study calculates, an HIV+ person has infected three others. But if treatment started shortly after people became infected, the model suggests, on average, the person would infect less than one person each, and the severe South African epidemic would die out in 14 years (see graph). "If we really put our mind to it, we can do things people previously didn't think were possible," says De Cock, noting that 3 million people in poor countries now receive ARVs that once were deemed impossibly expensive for them.

Geoffrey Garnett, an epidemiologist at Imperial College London, St. Mary's, co-authored an accompanying editorial that called the strategy "extremely radical" for pushing public health over individual benefits: It remains unclear that early treatment delays disease and death, and it could increase the risk of drug resistance developing. It also raises other ethical, financial, and logistical dilemmas. And, Garnett contends, eliminating the epidemic is too ambitious a goal. But even if the strategy didn't reach all infected people, it might have a big impact on spread, he says. "It's not an idea to dismiss out of hand, but it's really challenging," says Garnett. "The *Lancet* paper throws down the gauntlet to think about it."

Other research groups received little attention—or outright derision—when they published similar models a few years ago, but now, ARVs are less toxic and cheaper than ever, making the idea more realistic. Still, many question whether it could be pulled off. Today, an estimated 80% of the infected people in sub-Saharan Africa do not know their HIV status, and it's not simply a question of access to tests: In the United States, where testing is widely available, the Centers for Disease Control and Prevention says about 25% of infected people are unaware they are infected. Although the world has made much progress in delivering ARVs to poor countries, at the end of 2007, 6.7 million people who badly needed the drugs were not receiving them. Poor countries also typically use ARV regimens that are below the standards of those available in wealthy countries, which means toxicity and drug resistance remain serious issues for much of the world. In these countries, there are still staggering shortages of funds and trained people to test and treat. (Estimates suggest that by 2015, it will cost at least $41 billion annually to treat the 13.7 million HIV-infected people who then will have fewer than 200 CD4+s; wealthy countries currently donate some $10 billion.)



All together now. Although ARVs still fail to reach many people in countries such as Myanmar (*top*), a model based on South Africa (*bottom*) shows that immediate treatment of the entire HIV-infected population could stop the epidemic there.

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De Cock and his co-authors did a preliminary cost analysis for South Africa that showed by 2015, the country would have to spend three times as much on testing and treatment as it does today. But because of declining infection rates, costs would then begin a steady decline.

HIV/AIDS researcher Julio Montaner of the University of British Columbia, Vancouver, Canada, says he got "crucified" when he published a similar paper in *The Lancet* in 2006 and is "delighted" to see that WHO's "fancier" model supports his own. The key to the strategy, he contends, is cost: In the long run, it saves money. "Front-loading the investment on ARVs is a very wise investment, even in a time of financial crisis," says Montaner. But he agrees that more data are needed to prove that starting treatment at the earliest possible point does not lead to toxicities and drug resistance that offset the long-term benefits. "How would you like it if I say, 'You don't qualify for treatment, but I want to treat you so you don't expose anyone else?' "

De Cock says he would like researchers to launch small-scale trials of voluntary testing and immediate treatment to assess the impact on infected individuals and see whether it decreases the spread of HIV in populations, as the model predicts. WHO, he says, plans to hold a meeting early next year to discuss the strategy in more depth.

–JON COHEN